Group-Theoretic Discrimination of Twin Operation Spaces

From Non-Newtonian Calculus to the Class Equation and Weyl Integration

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Abstract

Generator-function arithmetics — non-Newtonian calculus, the multiplicative and bigeometric calculi, and the exponential hierarchy of operations — are classically presented as isolated constructions $x \oplus_\varphi y = \varphi^{-1}(\varphi(x) + \varphi(y))$. We place them in a single general theory: each is the *twin operation* $x \odot_{eg} y = g^{-1} \cdot \big((g \cdot x) \odot_e (g \cdot y)\big)$ for a monogenic group $G = \langle \varphi \rangle$, an instance of the operation space $\mathrm{Op}(E, \odot_e; G)$. We then *generalize* from monogenic $\mathbb{Z}$ to arbitrary symmetry groups and develop the invariants that *discriminate* among the resulting operations: the class equation for finite groups, primary decomposition for abelian groups, and Weyl integration for compact Lie groups. Along the way we correct the classification theorem of the literature: Op is the homogeneous space G/L, a group *iff* the structural kernel L is normal (computed counterexample: S_4 on $+$ mod 4). Finally we apply the generalizations and compare with the existing record: the discrimination of operation spaces into decision cells, the Weyl measure verified against random-matrix (CUE) statistics and Schur-character orthonormality, and the classical magma counts (n^n, q^{q+3}, $S_n \to 2$) recovered as special cases. Each statement is reproducible; the library carries 131 passing tests.

Keywords: non-Newtonian calculus, transport of structure, classification G/L, class equation, Weyl integration, Schur characters.
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1 Introduction

This paper follows a deliberate arc. We (1) recall the prior constructions of deformed arithmetic and what they contributed; (2) show that they all sit inside one general theory of *transport of structure*; (3) generalize that theory from cyclic to arbitrary symmetry groups; and (4) apply the generalization — discriminating operation spaces by group-theoretic invariants — comparing the outcomes with the existing record. The viewpoint is scientific and computational: every claim is matched to a reproducible calculation.

2 Antecedents and their contributions

Non-Newtonian and multiplicative calculus. Grossman and Katz [1] replaced the additive pair $(+, \cdot)$ by operations built from a generator φ,

$$x \oplus_\varphi y = \varphi^{-1}\big(\varphi(x) + \varphi(y)\big), \qquad x \otimes_\varphi y = \varphi^{-1}\big(\varphi(x) \cdot \varphi(y)\big), \tag{1}$$

yielding the multiplicative ($\varphi = \log$) and bigeometric calculi. Their contribution was a working family of calculi — derivatives, integrals, means — adapted to multiplicative phenomena, and the recognition that the "shape" of arithmetic is a modelling choice.

Transport of structure and deformation. In algebra, conjugating an operation by a map, $a \diamond_g b = g^{-1}(g(a) \diamond g(b))$, is the canonical equivariant deformation; its orbit theory is governed by the orbit–stabilizer and first-isomorphism theorems, and its obstructions by Gerstenhaber–Hochschild deformation cohomology [2]. This supplied the structural language but was rarely applied to *operations* as the deformed objects.

The exponential hierarchy. A recent foundations volume organizes (1) for $\varphi = \exp$ into a chain $+ \to \cdot \to x^{\ln y} \to \cdots$, the level-$n$ operation being $\exp^{(-n)}(\exp^{(n)} x + \exp^{(n)} y)$, and proposes a classification of the induced operations. Its contribution was to expose an infinite ladder of coherent arithmetics; its classification, however, over-claims a group structure, which we correct below.

Spectral side. On the eigenvalue side, Weyl's integration formula and the random-matrix theory of the classical compact groups [3, 4, 5] describe the distribution of conjugacy classes; these will provide both the tool and the benchmark for the Lie-group generalization.

3 The general theory and its corrected classification

Definition 1 (Transport)**.** Let a group G act on a set E with a binary operation $\odot_e$. The *twin operation* of $g \in G$ is

$$x \odot_{eg} y := g^{-1} \cdot ((g \cdot x) \odot_e (g \cdot y)). \tag{2}$$

Write $\pi : G \to \mathrm{Op}(E, \odot_e; G)$, $g \mapsto \odot_{eg}$, and let $L = \{g : \odot_{eg} = \odot_e\}$ be the *structural kernel.*

Every antecedent of §2 is the special case $G = \langle \varphi \rangle$ monogenic: (1) is (2) for the action $n \cdot x = \varphi^{(n)}(x)$, and the exponential hierarchy is $G = \langle \exp \rangle \cong \mathbb{Z}$. Thus non-Newtonian calculus and the hierarchy are *points* of a single object Op.

Proposition 2 (Kernel and fibres)**.** *L is a subgroup of G, and $\odot_{eg} = \odot_{eh} \iff hg^{-1} \in L \iff Lg = Lh$.*

Proof. Equality of (2) for g, h, after applying $h \cdot (-)$ and substituting $u = g \cdot x,\ v = g \cdot y$, reads $m \cdot (u \odot_e v) = (m \cdot u) \odot_e (m \cdot v)$ with $m = hg^{-1}$, i.e. $m \in L$. □

Corollary 3 (Classification)**.** *$Lg \mapsto \odot_{eg}$ is a bijection $L \backslash G \to \mathrm{Op}$; hence $|\mathrm{Op}| = [G : L]$ and Op is a homogeneous G-set.*

Remark 4 (What it means)*.* The operation space is not free data: it is rigidly the coset space $L \backslash G$, so its size and structure are fixed by the pair (G, L) alone. Choosing a twin operation is choosing a coset — nothing more, nothing less.

Theorem 5 (Group structure $\iff$ normality)**.** *$\odot_{eg} * \odot_{eh} := \odot_{e,gh}$ is well defined (so $\mathrm{Op} \cong G/L$ is a group and π a homomorphism) if and only if $L \trianglelefteq G$.*

Proof. Replacing g, h by ag, bh $(a, b \in L)$ preserves the product coset for all $b \in L$ iff $g^{-1}Lg = L$ for all g. □

Remark 6 (Correction to the literature)*.* The foundations volume asserts π is always a homomorphism, hence $L \trianglelefteq G$. This fails: for $G = S_4$ on $+$ mod 4 the kernel has order 2 and is *not* normal, so Op ($|\mathrm{Op}| = 12$) is a homogeneous space, not a group. Corollary 3 is the correct universal statement; the group case is Theorem 5.

4 Generalization: discrimination by group invariants

Leaving the monogenic case, we equip the decision problem "choose $\arg\min_{\mathrm{Op}} J$" with the discriminating invariants of G.

4.1 Finite groups: the class equation

Definition 7. A cost J on a finite group is a *class function* if $J(wxw^{-1}) = J(x)$.

Proposition 8 (Class-function reduction)**.** *If $L \trianglelefteq G$ and J is a class function on* $\mathrm{Op} = G/L$*, then J is constant on conjugacy classes, and the class equation*

$$|\mathrm{Op}| = |Z(\mathrm{Op})| + \sum_i [\mathrm{Op} : C_{\mathrm{Op}}(x_i)] \tag{3}$$

enumerates the decision cells.

Theorem 9 (Normalizer orbits)**.** *The relabeling $\odot_{eg} \mapsto \odot_{e,wgw^{-1}}$ is well defined on* Op *exactly for $w \in \mathrm{N}_G(L)$; its orbits are the decision cells (Burnside), reducing to the conjugacy classes of* Op *when $L \trianglelefteq G$.*

Remark 10 (What it means). The quotient $\mathrm{N}_G(L)/L$ is the *Weyl group of the configuration*: the largest symmetry that acts on Op through the kernel. Operations in the same orbit are interchangeable under a symmetry already present in the system, so *no* structural criterion can separate them — the orbits are therefore the genuine units of decision.

The class equation is a *symmetry spectrum* of the decision space (Figure 1, Table 1): the centre collects robust, frame-canonical operations; large classes are generic; small classes are highly symmetric.

Table 1: Class equations of representative groups (computed).

Group	$\lvert G\rvert$	#classes	class equation
S_3	6	3	$6 = 1 + 2 + 3$
D_4	8	5	$8 = 2 + 2 + 2 + 2$
Q_8	8	5	$8 = 2 + 2 + 2 + 2$
S_4	24	5	$24 = 1 + 3 + 6 + 6 + 8$

Reading. A spectrum of many singleton classes (the abelian case, all gold) means every operation is its own decision cell — maximal discrimination. The wide blocks of S_4 mean many operations collapse into one cell. That D_4 and Q_8 share a spectrum warns that the class equation is a coarse, though exactly computable, invariant: it bounds the decision granularity without fully resolving the group.

4.2 Abelian groups: primary decomposition

For abelian G the kernel is normal and $\mathrm{Op} = G/L$ factors along its Sylow-p components.

Theorem 11 (Primary decomposition)**.** $G \cong \prod_p G_p$*,* $L = \prod_p L_p$*, and* $\mathrm{Op} \cong \prod_p G_p/L_p = \prod_p \mathrm{Op}_p$*.*

Hence operations factor uniquely into p-primary parts, $|\mathrm{Op}| = \prod_p |\mathrm{Op}_p|$, and a separable cost is optimized componentwise — a Chinese Remainder factorization of the decision (Table 2).

Remark 12 (What it means). A decision over Op is equivalent to independent decisions over its p-primary factors. This turns a product-sized search ($\prod_p |\mathrm{Op}_p|$) into a sum-sized one ($\sum_p |\mathrm{Op}_p|$) — the operation-level analogue of the discrete Fourier / CRT speed-up.

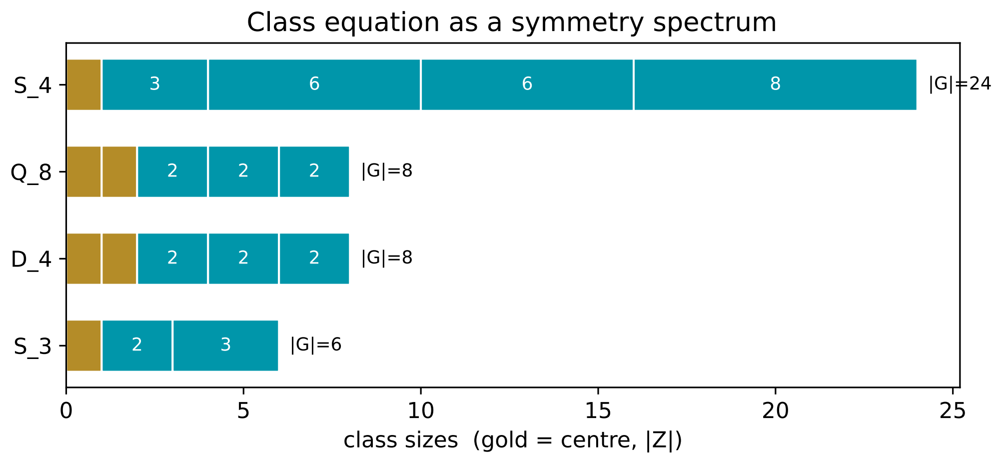

Figure 1: The class equation as a symmetry spectrum: conjugacy-class sizes partition $|G|$ (gold = centre). D_4 and Q_8 share $8 = 2 + 2 + 2 + 2$.

Table 2: Primary decomposition of $\mathrm{Op} = C_{12}/\langle 6\rangle$ (computed).

prime p	$\lvert G_p\rvert$	$\lvert L_p\rvert$	$\lvert\mathrm{Op}_p\rvert = [G_p : L_p]$
2	4	2	2
3	3	1	3
product	12	2	$6 = [G : L]$

Reading. The size-6 decision space $C_{12}/\langle 6\rangle$ *is* $\mathrm{Op}_2 \times \mathrm{Op}_3$ (2×3); the kernel itself splits as $L_2 \times L_3$. One never decides over the whole space, only over a 2-element and a 3-element factor.

For a compact connected G the class equation becomes the Weyl integration formula $\int_G J\,d\mu = \frac{1}{|W|}\int_T J(t)|\Delta(t)|^2 dt$. For $SU(2)$ the class angle carries $(2/\pi)\sin^2\theta\,d\theta$; for $U(n)$ the eigen-phase density is the squared Vandermonde

$$d\mu(\theta) = \frac{1}{n!\,(2\pi)^n}\prod_{j<k}\left|e^{i\theta_j} - e^{i\theta_k}\right|^2 d\theta, \tag{4}$$

the CUE density, with irreducible characters the Schur polynomials $\chi_\lambda = s_\lambda(e^{i\theta})$.

Remark 13 (The validated $SU(2)$ law is the $n = 2$ Vandermonde)*.* The class density used in §5 is not a separate input: it is (4) at $n = 2$. An $SU(2)$ matrix has eigenvalues $e^{\pm i\theta}$, so the squared Vandermonde is

$$\left|e^{i\theta} - e^{-i\theta}\right|^2 = \left|2i\sin\theta\right|^2 = 4\sin^2\theta,$$

and normalising it to a probability density on the class angle $\theta \in [0, \pi]$ — where $\int_0^\pi 4\sin^2\theta\,d\theta = 2\pi$ — gives exactly $\frac{2}{\pi}\sin^2\theta$. The repulsion factor $\sin^2\theta$, i.e. eigenvalues avoiding coincidence, is the level repulsion confirmed in Figure 3. The finite class equation (3) and the Lie density (4) are thus one statement — orbit volume = index/stabiliser — read in the discrete and the continuous regime.

5 Applications and comparison with the existing record

5.1 Discrimination of operation spaces

The generalization turns a large operation space into a few decision cells. For S_4 on $+ \bmod 4$ the 12 twin operations split into 5 cells of sizes $\{1, 1, 2, 4, 4\}$ under the Weyl group $\mathrm{N}_G(L)/L$ of order 2 (Table 3); the two singletons are the canonical operations. Figure 2 shows the reduction across several pairs. *Comparison:* where the prior classification would (incorrectly) read Op as a group, the corrected theory reads it as a homogeneous space and supplies its genuine decision structure.

Table 3: Decision data for S_4 on $+$ mod 4 (computed): L non-normal, Op a homogeneous space.

quantity	value
$\lvert G\rvert$, $\lvert L\rvert$, normal?	24, 2, no
$\lvert\mathrm{Op}\rvert = [G : L]$	12
$\lvert \mathrm{N}_G(L)\rvert$, Weyl $\lvert \mathrm{N}_G(L)/L\rvert$	4, 2
decision cells (sizes)	5 $(\{1, 1, 2, 4, 4\})$

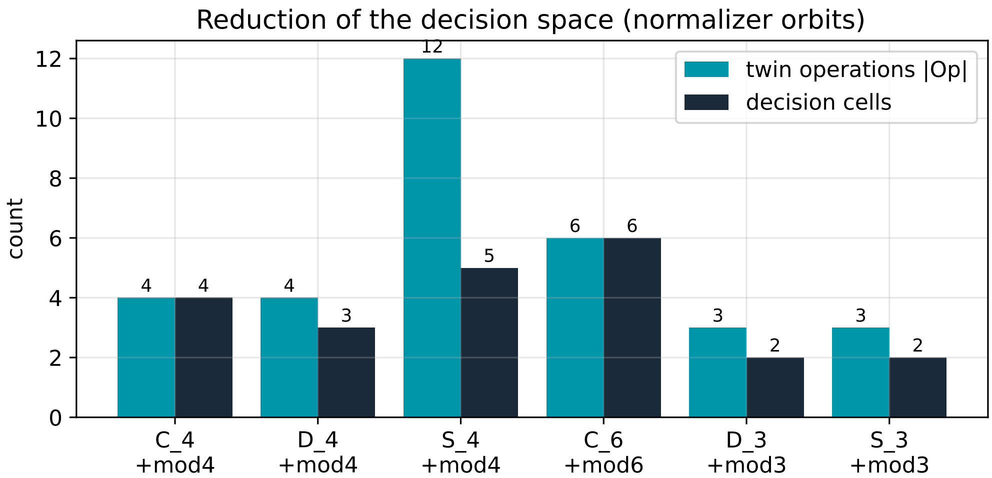

Figure 2: Reduction of the decision space: distinct twin operations $\lvert\mathrm{Op}\rvert = [G : L]$ vs decision cells (orbits of $\mathrm{N}_G(L)$), for finite $(G, + \bmod m)$ pairs.

Reading. The gap between the two bars is the dividend of the theory: for S_4 a 12-way choice becomes a 5-way one. The gap is widest for the richest non-abelian groups, where conjugation identifies the most operations.

5.2 The Weyl measure against random-matrix statistics

The Lie generalization is validated against the existing spectral record. The $SU(2)$ class density matches $(2/\pi)\sin^2\theta$ (Figure 3, left); the CUE moments $\mathbb{E}[\mathrm{tr}\, U] = 0$, $\mathbb{E}\lvert\mathrm{tr}\, U\rvert^2 = 1$ are recovered for $U(2), U(3), U(4)$ (Figure 3, right; Table 4); and the measure (4) is pinned down by Schur-character orthonormality $\langle \chi_\lambda, \chi_\mu\rangle = \delta_{\lambda\mu}$ (Figure 4). *Comparison:* these reproduce classical random-matrix and representation-theoretic facts [4, 6], confirming that the decision measure on qudit gate classes is the correct one.

Table 4: CUE moments under the Weyl measure (Haar Monte-Carlo, 2×10^4 samples).

	$U(2)$	$U(3)$	$U(4)$
$\mathbb{E}[\mathrm{tr}\, U]$ (target 0)	≈ 0.00	≈ 0.00	≈ 0.01
$\mathbb{E}\lvert\mathrm{tr}\, U\rvert^2$ (target 1)	1.01	0.99	1.00

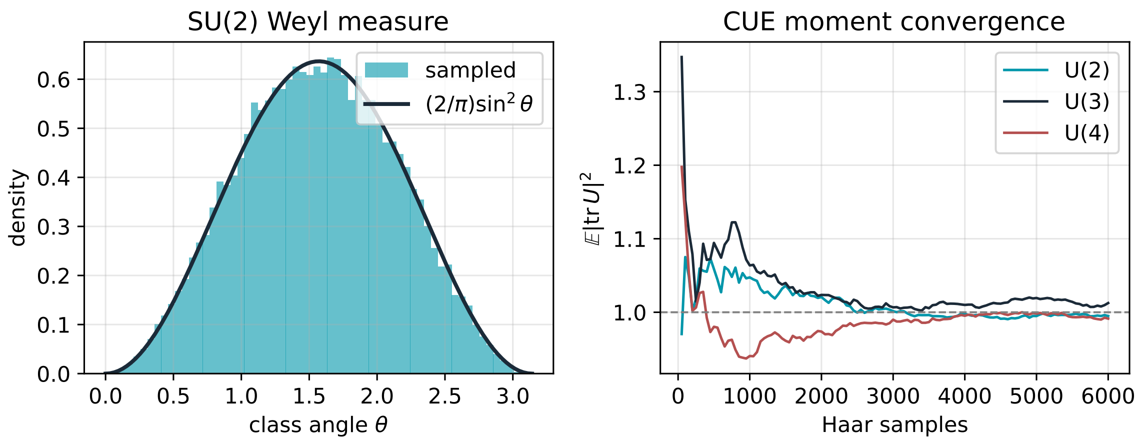

Figure 3: Left: $SU(2)$ class-angle density, sampled vs $(2/\pi)\sin^2\theta$. Right: convergence of $\mathbb{E}|\mathrm{tr}\,U|^2 \to 1$ for $U(2), U(3), U(4)$.

Reading. The left panel is a direct, visual confirmation that our class measure on $SU(2)$ is the Weyl law, not an artefact; the right panel shows the same for higher rank through a moment that the measure must return as exactly 1. Agreement here is what licenses decisions taken against (4).

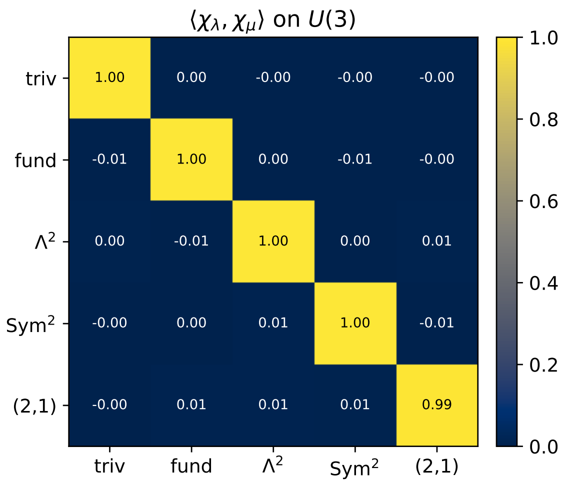

Figure 4: Computed Gram matrix $\langle\chi_\lambda, \chi_\mu\rangle$ of Schur characters on $U(3)$: the identity, confirming the Weyl measure (4).

Reading. Orthonormality is the sharpest possible check: the characters form an orthonormal basis of class functions *only* under the correct measure, so a Gram matrix indistinguishable from the identity pins down (4) uniquely. The off-diagonal noise ($\sim 10^{-2}$) is the Monte-Carlo error, not a systematic deviation.

5.3 Classical magma counts as special cases

Finally, the orbit machinery reproduces the existing enumeration of G-invariant magmas, $\prod_O |\mathrm{Fix}_E(\mathrm{Stab}_G(O))|$: the cyclic count $C_n \mapsto n^n$, the symmetric collapse S_n $(n \geq 4) \mapsto 2$, and the linear-group stabilization $\mathrm{GL}_n(\mathbb{F}_q) \mapsto q^{q+3}$ (computed for small cases). *Comparison:* these classical counts emerge as a by-product of the same decision-cell decomposition. Table 5 summarizes the arc.
Reading. Read left to right, each row is the arc in miniature: a known construction, its home in

Table 5: The four-movement arc: antecedents, their place, their generalization, and the new outcome.

antecedent	as twin (G)	generalizes to	new outcome
multiplicative calculus [1]	$\langle\log\rangle$	arbitrary G	class-equation discrimination
exponential hierarchy	$\langle\exp\rangle \cong \mathbb{Z}$	finite / abelian / Lie	corrected classification (homog. space)
CUE eigenvalue statistics [4]	conjugation by $U(n)$	qudit class costs	Weyl-verified decision measure
invariant-magma counts	finite G on E^2	normalizer orbits	decision-cell reduction

the general theory, the direction it generalizes, and the new result that follows. The framework's value is that the same final column is produced by one machine for every row.

All results are realized in `arithmetics.symbolic.twin` (SymPy, NumPy, mpmath): exact finite-group computation (conjugacy, centralizers, centre, normality, quotients), exact one-parameter analysis for $SU(2), SO(2)$, Haar Monte-Carlo for $SO(3), SU(n)$, and Schur characters by the bialternant. The figures and tables are regenerated by `papers/make_figures.py`; each theorem is paired with a unit test (131 tests pass).

6 Discussion

The arc closes a gap: classical generator-function arithmetics are unified as twin operations, generalized from $\mathbb{Z}$ to arbitrary symmetry, and discriminated by a single family of invariants that simultaneously recovers known counts, matches random-matrix statistics, and corrects a normality error. The honest limitation is structural: by equivariance every twin operation is isomorphic to the base, so the discriminating invariants are conjugation-level (the class equation) or frame-dependent — the latter is the subject of the companion, technological paper.

A Complete proofs

Theorem 9 (normalizer orbits). We first show the relabeling $r_w : \odot_{eg} \mapsto \odot_{e,wgw^{-1}}$ is well defined on Op iff $w \in \mathrm{N}_G(L)$. By Proposition 2 $\odot_{eg}$ determines, and is determined by, the right coset Lg. For $a \in L$, $w(ag)w^{-1} = (waw^{-1})(wgw^{-1})$, so $L\,w(ag)w^{-1} = L\,wgw^{-1}$ for all $a \in L$ iff $waw^{-1} \in L$ for all $a \in L$, i.e. iff $wLw^{-1} = L$. Hence r_w is a well-defined map Op $\to$ Op exactly for $w \in \mathrm{N}_G(L)$, and $w \mapsto r_w$ is a group action of $\mathrm{N}_G(L)$ (since $r_w r_{w'} = r_{ww'}$ on cosets). The decision cells are its orbits; by Burnside their number is $\frac{1}{|\mathrm{N}_G(L)|}\sum_w |\mathrm{Fix}(r_w)|$. If $L \trianglelefteq G$ then $\mathrm{N}_G(L) = G$ and r_w is conjugation on G/L, whose orbits are the conjugacy classes. □

Theorem 11 (primary decomposition). Let $|G| = \prod_i p_i^{a_i}$ and $G_p = \{g : \mathrm{ord}(g) = p^k\}$. For each p put $q = |G|/p^a$ and choose, by Bézout, u with $uq \equiv 1 \pmod{p^a}$; set $e_p = uq$, so $e_p \equiv 1 \pmod{p^a}$ and $e_p \equiv 0 \pmod{p'^{a'}}$ for $p' \neq p$. Then $\pi_p(g) := g^{e_p} \in G_p$ and $\sum_p e_p \equiv 1 \pmod{|G|}$, whence $\prod_p \pi_p(g) = g$; this realises $G \cong \prod_p G_p$. Any subgroup $L \leq G$ is a finite abelian group, so $L = \prod_p (L \cap G_p) = \prod_p L_p$. Quotienting factorwise gives $G/L \cong \prod_p G_p/L_p$. □

Proposition 8 (class-function reduction). If J is a class function and x, x' are conjugate, $J(x) = J(x')$; thus J is constant on each conjugacy class and $\arg\min J$, being defined by a value of J, is a union of classes. Evaluating one representative per class therefore suffices, and (3) counts the classes with multiplicities. □

Counting formula. A G-invariant magma is a G-equivariant map $\odot : E \times E \to E$. Equivariance fixes $\odot$ on a whole orbit $O \subseteq E \times E$ once its value at a representative (i, j) is chosen; that value must lie in $\mathrm{Fix}_E(\mathrm{Stab}_G(i, j))$. Independent choices over orbits give $\prod_O |\mathrm{Fix}_E(\mathrm{Stab}_G(O))|$. □

B Worked computation: S_4 on $+ \bmod 4$

The structural kernel is $L = \{e, (1\,3)\}$, of order 2. Its normalizer is $\mathrm{N}_{S_4}(L) = \{e, (1\,3), (0\,2), (0\,2)(1\,3)\}$ of order 4 (the centraliser of the involution $(1\,3)$), so the Weyl group $\mathrm{N}_G(L)/L$ has order 2, generated by the image of $(0\,2)$. The conjugacy classes of S_4 have sizes $1, 6, 3, 8, 6$ (identity, transpositions, double transpositions, 3-cycles, 4-cycles). The $\mathrm{N}_G(L)$-action on the $[G : L] = 12$ operations has the orbit profile $\{1, 1, 2, 4, 4\}$: two fixed operations (the canonical, robust choices), one 2-orbit, and two 4-orbits. A purely structural criterion thus reduces the 12-way choice to 5 cells, ranked by symmetry.

C Finite discovery table

Table 6 lists the computed classification across a catalogue of finite groups and two base operations ($+ \bmod m$ and the rigid left projection $x \odot y = x$). The left projection is preserved by every permutation ($L = G$, one cell); the additions are non-trivial, and L is non-normal exactly for the non-abelian groups, where $|\mathrm{Op}| = [G : L]$ exceeds the cell count.

Table 6: Computed twin-operation classification (decision cells via $\mathrm{N}_G(L)$-orbits).

Group	base	$\lvert G\rvert$	$\lvert L\rvert$	$[G : L]$	$L \trianglelefteq G$	cells
C_2	$+ \bmod 2$	2	1	2	yes	2
C_3	$+ \bmod 3$	3	1	3	yes	3
C_4	$+ \bmod 4$	4	1	4	yes	4
V_4	$+ \bmod 4$	4	1	4	yes	4
D_3	$+ \bmod 3$	6	2	3	no	2
D_4	$+ \bmod 4$	8	2	4	no	3
S_3	$+ \bmod 3$	6	2	3	no	2
S_4	$+ \bmod 4$	24	2	12	no	5
any	left-proj	$\lvert G\rvert$	$\lvert G\rvert$	1	yes	1

Reading. The table separates three regimes at a glance: *rigid* (left projection, one cell, $L = G$); *abelian non-trivial* (L trivial and normal, Op a group of size $[G : L]$); and *non-abelian* (L non-normal, so Op is only a homogeneous space and the cell count drops below $[G : L]$). The last regime is exactly where the correction of Remark 6 bites.

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